Lec 2- Microbial Structure

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Characteristics of Cells

Eukaryotic cells: animals, plants, fungi, and protists

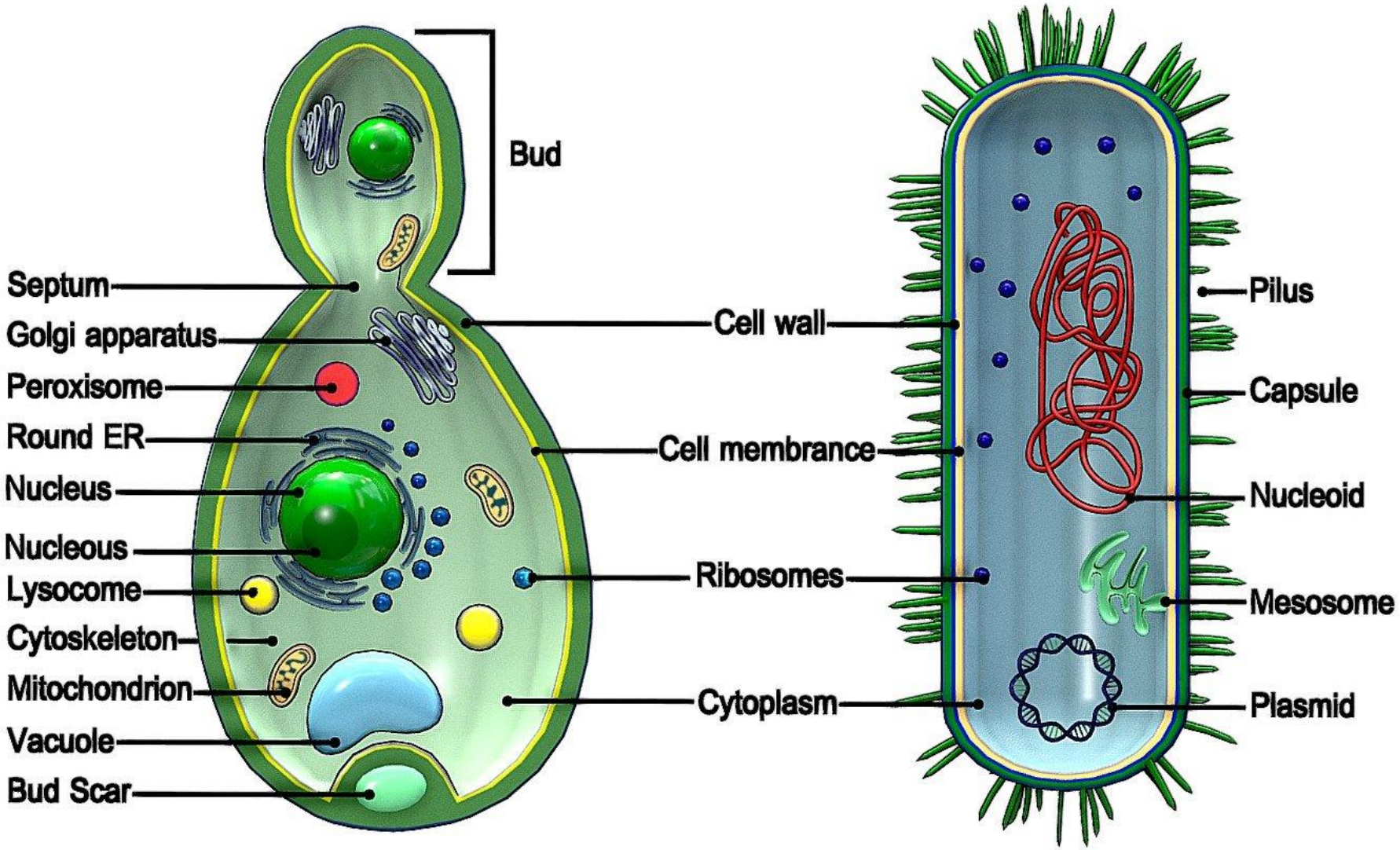
- Contain membrane-bound **organelles** that compartmentalize the cytoplasm and perform specific functions
- Contain double-membrane bound **nucleus** with DNA chromosomes

Prokaryotic cells: bacteria and archaea

- No nucleus or other membrane-bound organelles

Fungal cell

Bacterial cell



Prokaryotic Profiles

Glycocalyx-surface coating

- Coating of molecules external to the cell wall, made of sugars and/or proteins
- Two types:
  1. Slime layer - loosely organized and attached
  2. Capsule - highly organized, tightly attached
- Functions:
  - Protect cells from dehydration and nutrient loss
  - Inhibit killing by white blood cells by phagocytosis, contributing to pathogenicity
  - Attachment - formation of **biofilms**

Eukaryotic Profiles

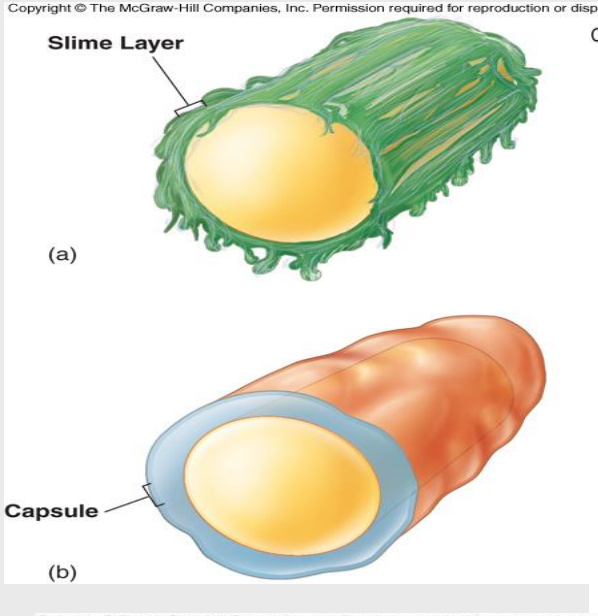
Mucilaginous substances

Fungi can secrete mucilaginous substances made of sugars, where these substances accumulate on the outer surface of the cell wall, forming an extracellular matrix,

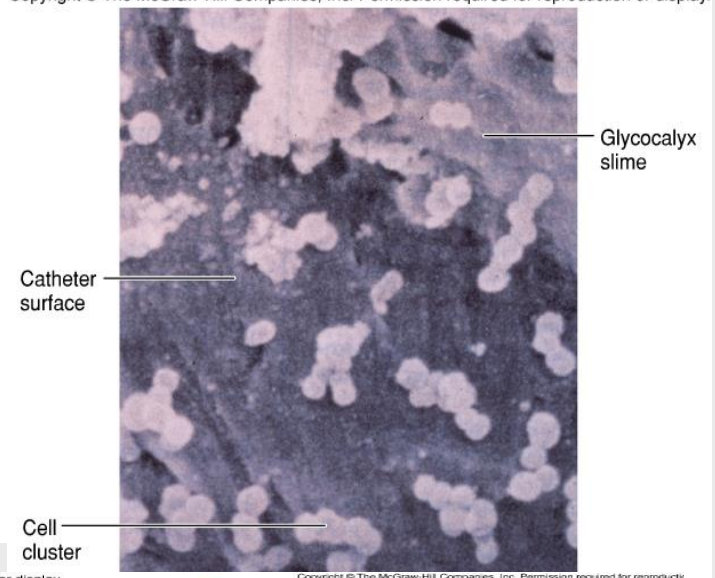
the extracellular matrix, which plays an important role in the adhesion of fungal structures to the surfaces growing on them. Where the fungus sticks to the surfaces and does not wash away with the currents of water.

Prokaryotic Profiles

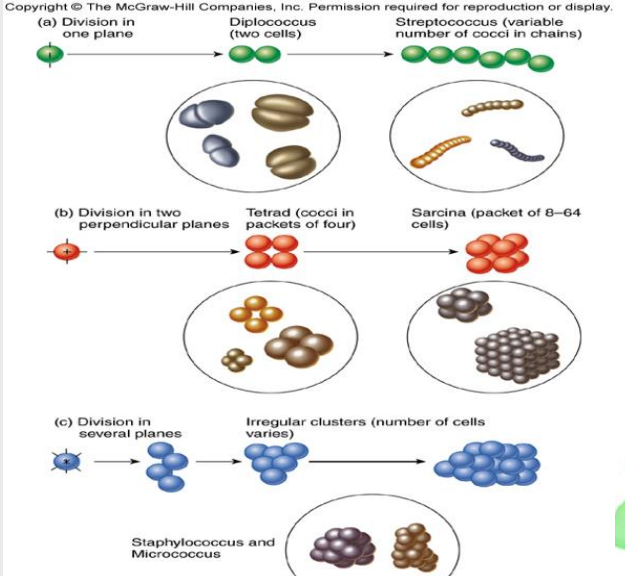
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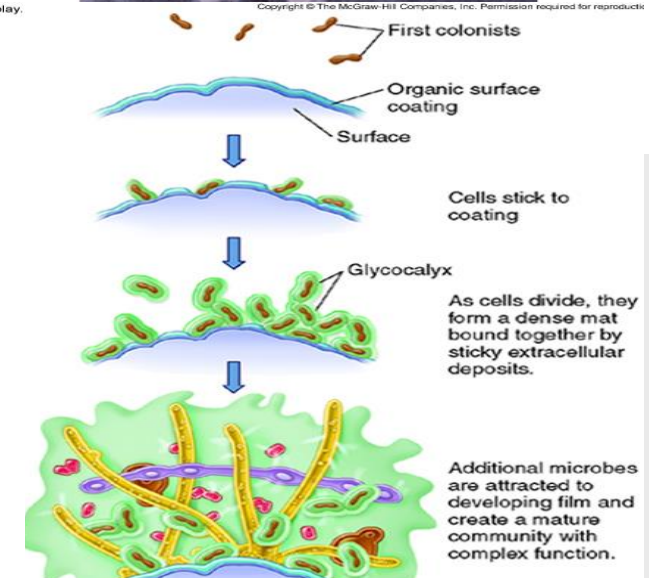


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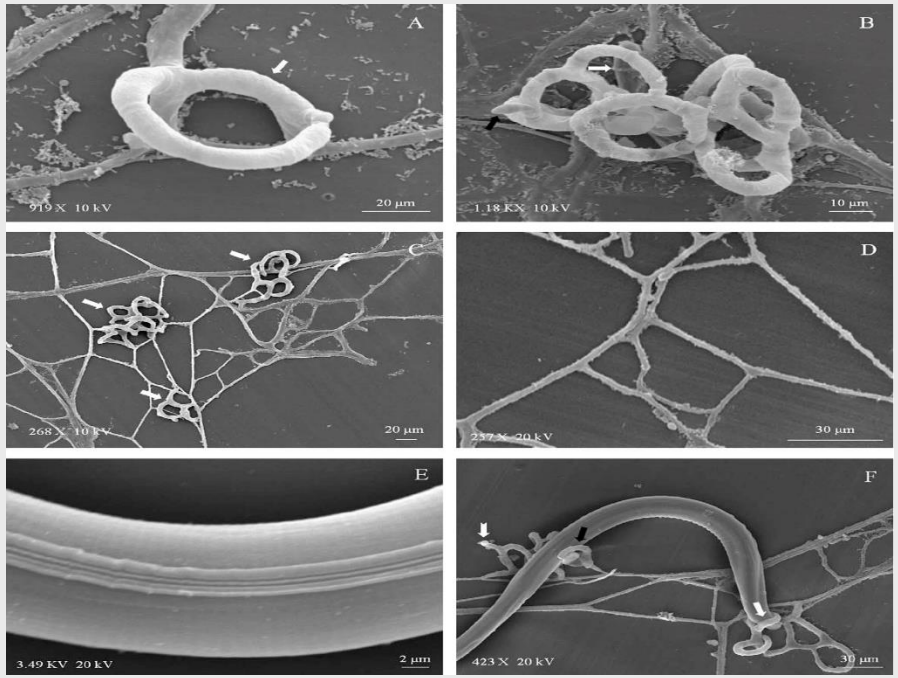
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Eukaryotic Profiles





Prokaryotic Profiles

Eukaryotic Profiles

Structure of Cell Walls

- Determines cell shape, prevents **lysis** (bursting) or collapsing due to changing osmotic pressures
- **Peptidoglycan** is primary component:
  - Unique macromolecule composed of a repeating framework of long glycan chains cross-linked by short peptide fragments
  - Includes **teichoic acid** and **lipoteichoic acid**: function in cell wall maintenance and enlargement during cell division; move cations across the cell envelope; stimulate a specific immune response
  - Outermost layer contains **lipopolysaccharides** and **lipoproteins (LPS)**
    - Lipid portion (endotoxin) may become toxic when released during infections
    - May function as receptors and blocking immune response
    - Contain **porin** proteins in upper layer – regulate molecules entering and leaving cell

- Most fungal cells are surrounded by a rigid cell wall - protecting them from the effects of osmotic stresses. This wall is a microfiber .**Fungal cell walls are rigid and contain complex polysaccharides called chitin and glucans.**
- **Ergosterol is the steroid molecule** in the cell membranes that replaces the cholesterol found in animal cell membranes. It helps in maintaining cell membrane structure, function, fluidity, and permeability

Prokaryotic Profiles

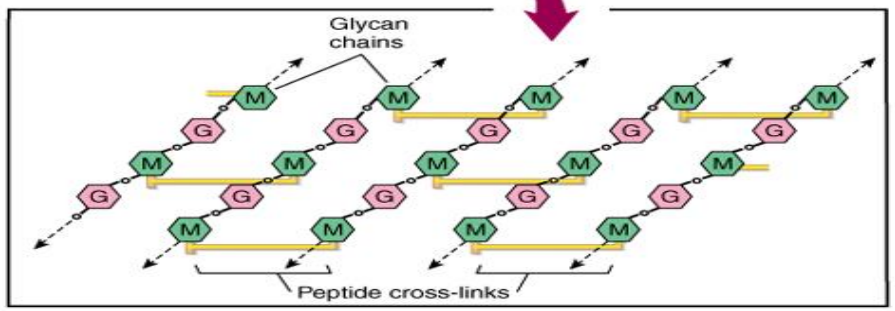
Peptidoglycan

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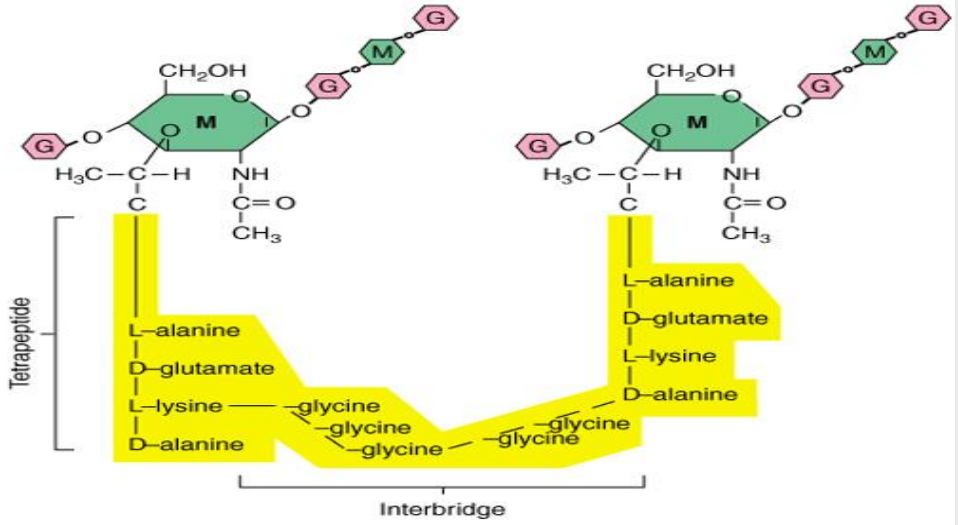
(a) The peptidoglycan of a cell wall forms a pattern similar to a chain-link fence. It is actually one giant molecule that surrounds and supports the cell.



(b) This shows the molecular pattern of peptidoglycan. It has alternating glycans (G and M) bound together in long strands. The G stands for *N*-acetyl glucosamine, and the M stands for *N*-acetyl muramic acid. Adjacent muramic acid molecules on parallel chains are bound by a cross-linkage of peptides (yellow).

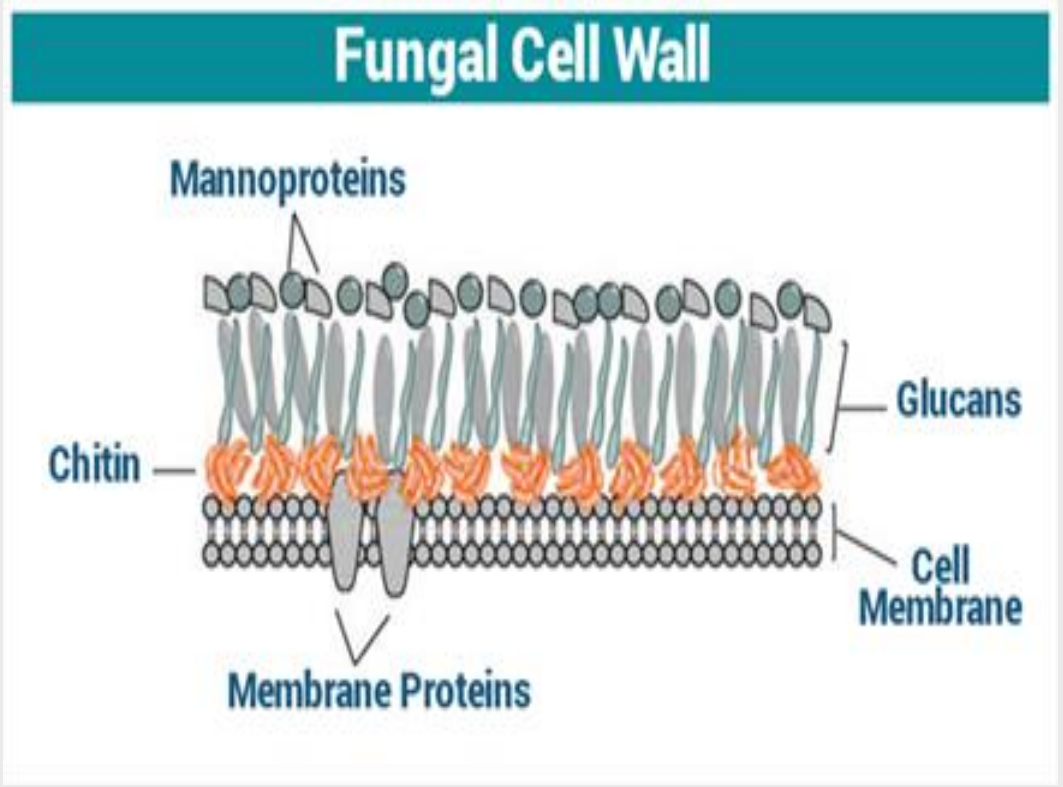


(c) An enlarged view of the links between the muramic acids. Tetrapeptide chains branching off the muramic acids connect by amino acid interbridges. The amino acids in the interbridge can vary or may be lacking entirely. It is this linkage that provides rigid yet flexible support to the cell.

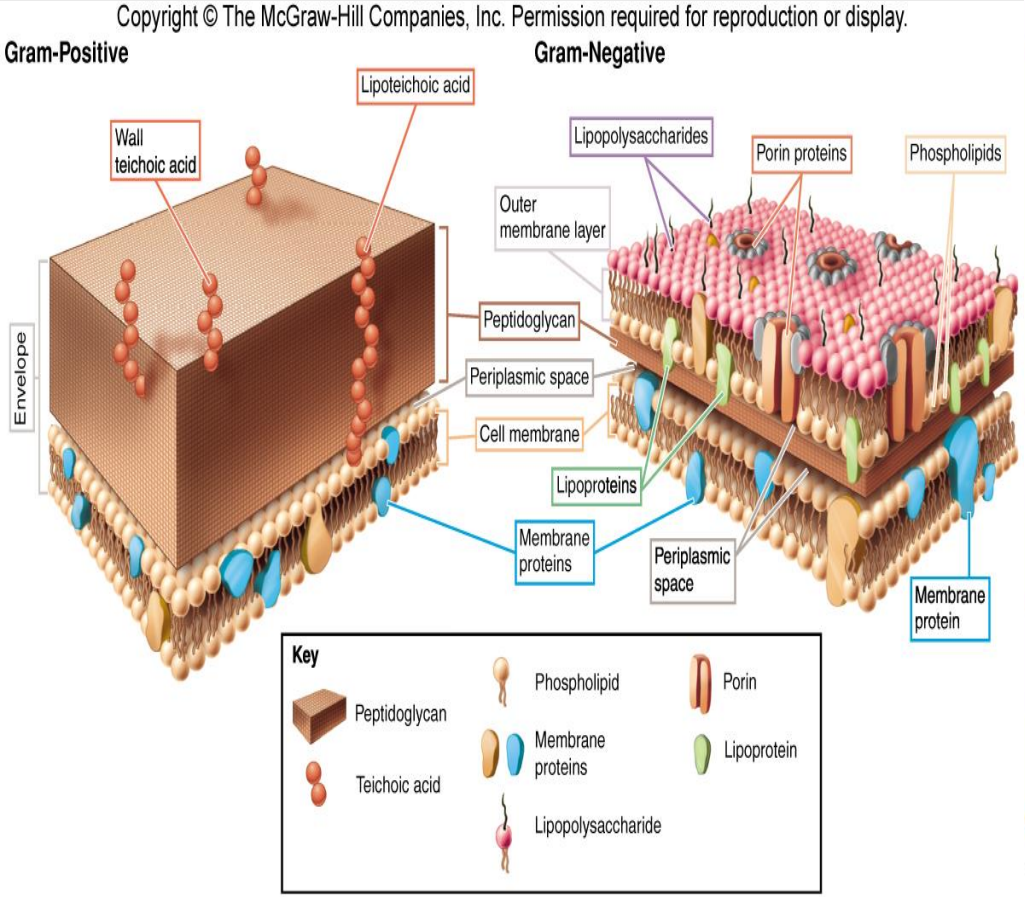


Eukaryotic Profiles

No Peptidoglycan



Comparison of Gram-Positive and Gram-Negative

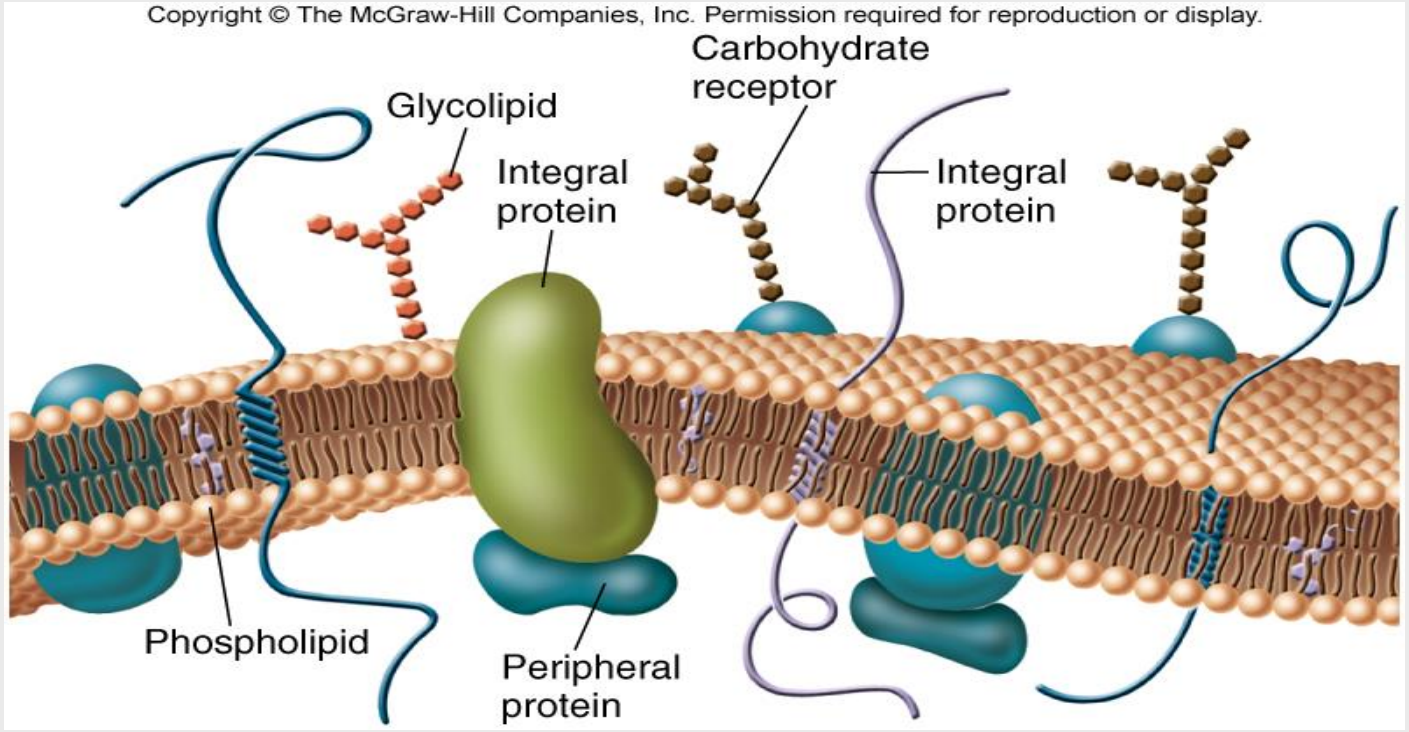


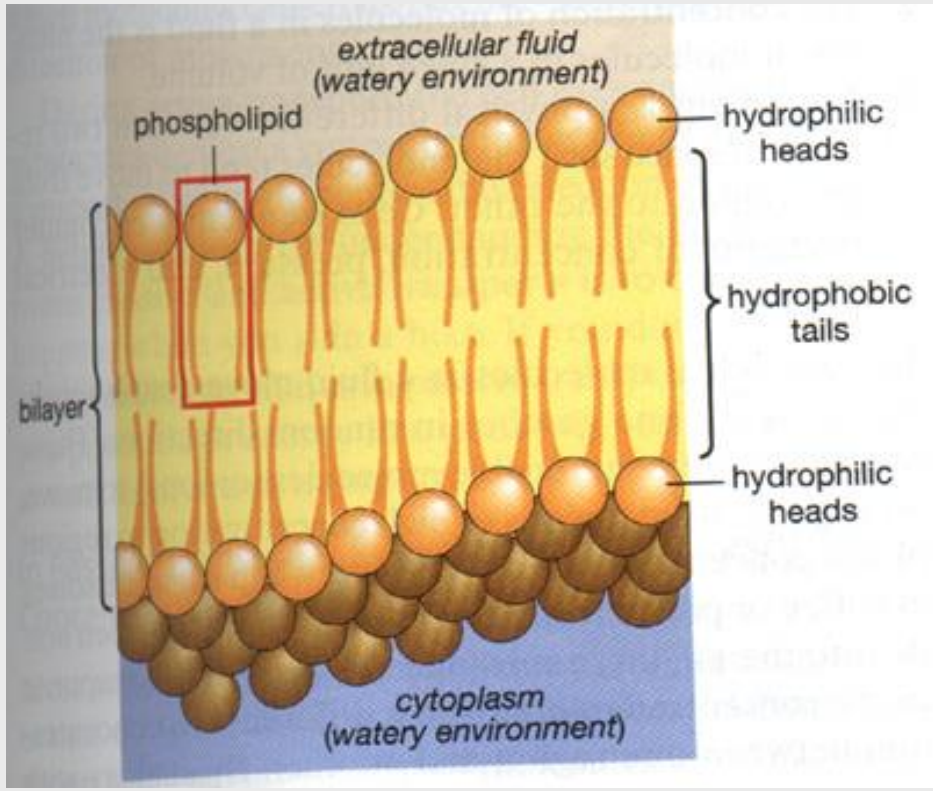
Characteristic	Gram-Positive	Gram-Negative
Number of major layers	One	Two
Chemical composition	Peptidoglycan Teichoic acid Lipoteichoic acid Mycolic acids and polysaccharides*	Lipopolysaccharide (LPS) Lipoprotein Peptidoglycan Porin proteins
Overall thickness	Thicker (20–80 nm)	Thinner (8–11 nm)
Outer membrane	No	Yes
Periplasmic space	Narrow	Extensive
Permeability to molecules	More penetrable	Less penetrable

*In some cells.

Cell Membrane Structure

- Phospholipid bilayer with embedded proteins – **Fluid mosaic model**
- Functions in:
  - Providing site for energy reactions, nutrient processing, and synthesis
  - Passage of nutrients into the cell and the discharge of wastes
  - Cell membrane is selectively permeable





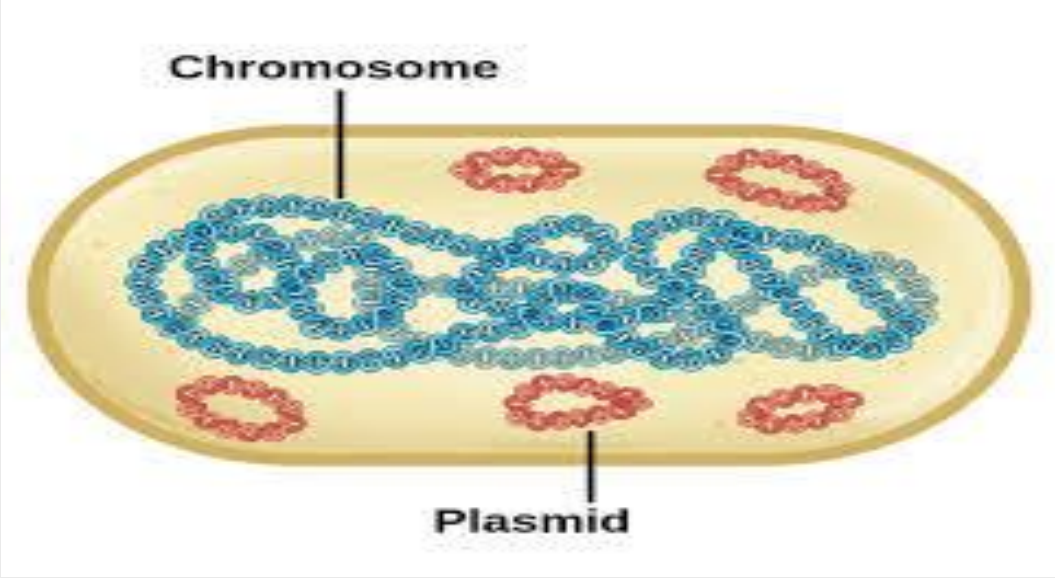
Membrane Lipids

Microbial Internal Structures

- Cell cytoplasm:
  - Dense gelatinous solution of sugars, amino acids, and salts
  - 70-80% water
    - Serves as solvent for materials used in all cell functions

Prokaryotic Profiles

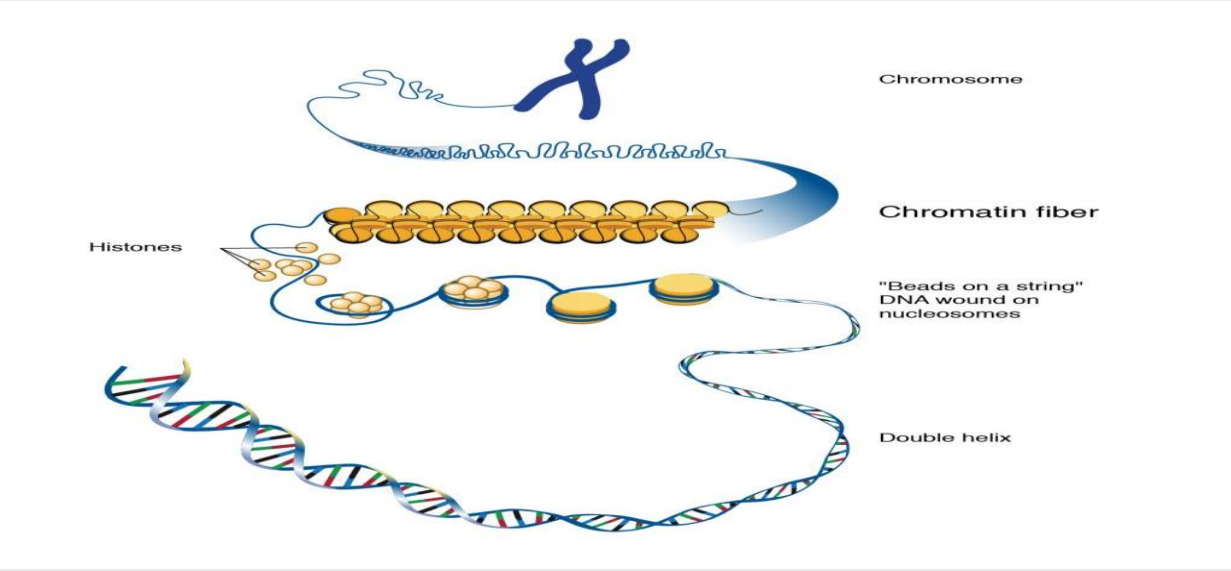
- Chromosome
- Single, circular, double-stranded DNA molecule that contains all the genetic information required by a cell
- Aggregated in a dense area called the **nucleoid**
- DNA is tightly coiled



Eukaryotic Profiles

A true nucleus is clear and of an orderly structure, as in other eukaryotic organisms. It is a spherical body consisting of: a nuclear envelope (consists of two layers - a nuclear fluid - the nucleolus, which is the control center for all vital processes in the cell.

Histones



Nucleic acids consist of chains of chemical units called nucleotides. Each nucleotide consists of three main components:

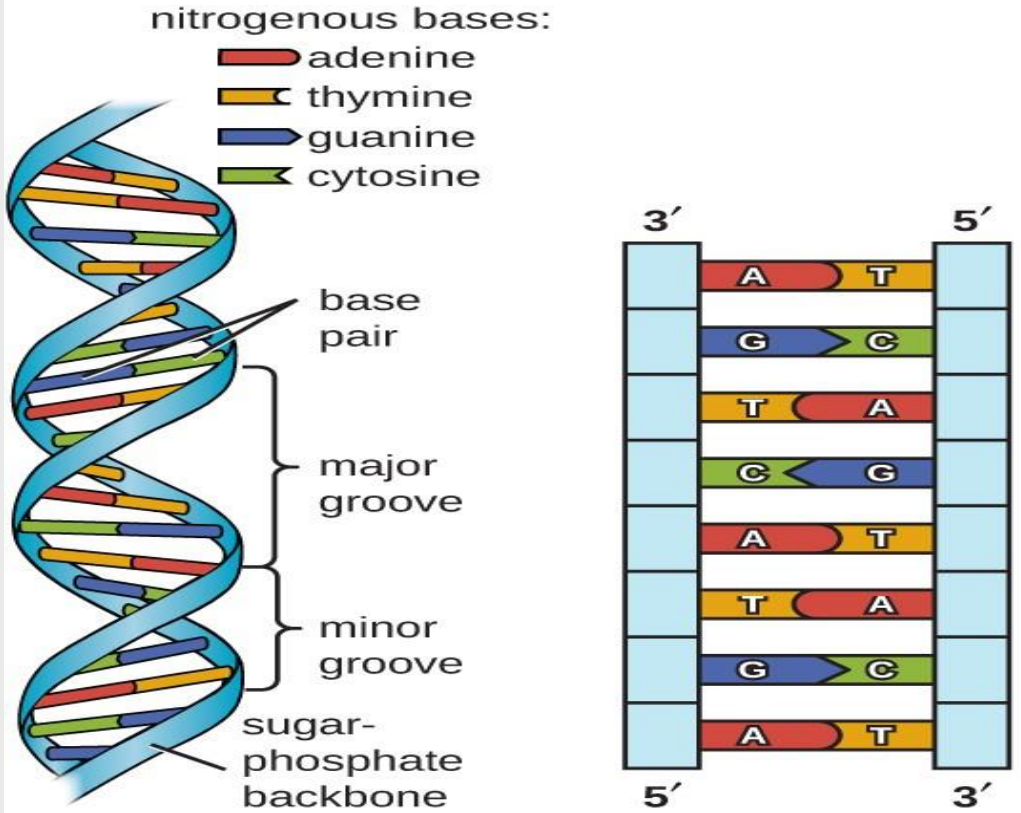
1.A pentaglyceride (ribose, or deoxyribose) molecule.

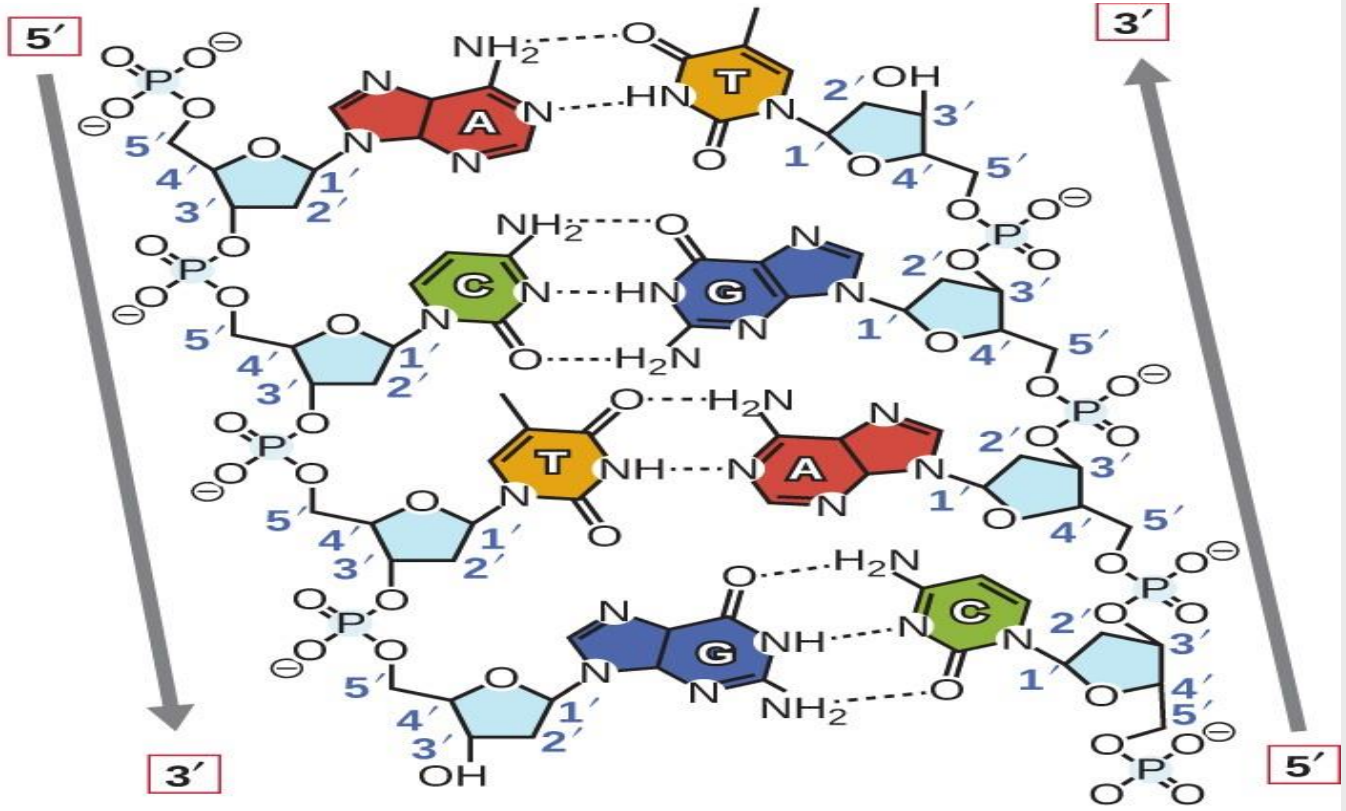
2. A group of phosphates.

3. Nitrogenous base. Nitrogenous bases consist of:

A. Purines : they include two bases: adenine A and guanine G, each consisting of two rings.

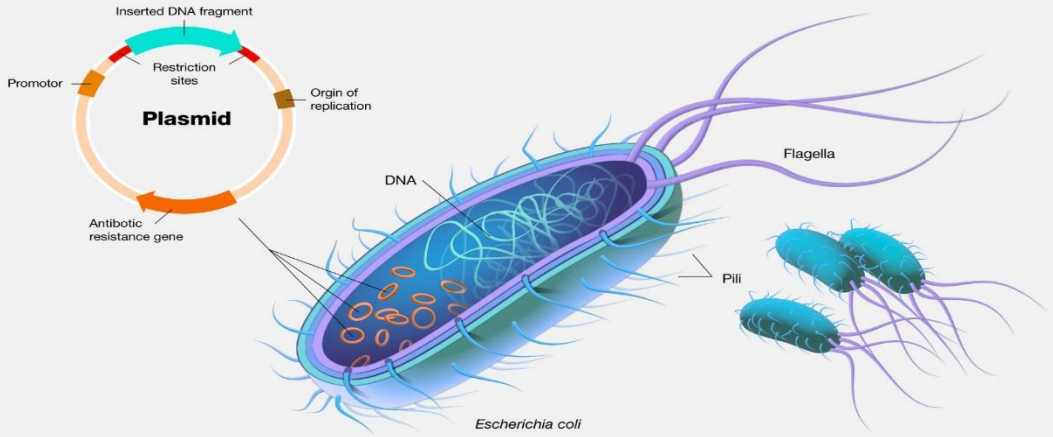
B. Pyrimidines: they contain three bases: thymine T, cytosine C, and uracil U, each of which consists of one ring.





Prokaryotic Profiles

- Plasmids
  - Small circular, double-stranded DNA
  - Free or integrated into the chromosome
  - Duplicated and passed on to offspring
  - Not essential to bacterial growth and metabolism
  - May encode antibiotic resistance, tolerance to toxic metals, enzymes, and toxins
  - Used in genetic engineering - readily manipulated and transferred from cell to cell



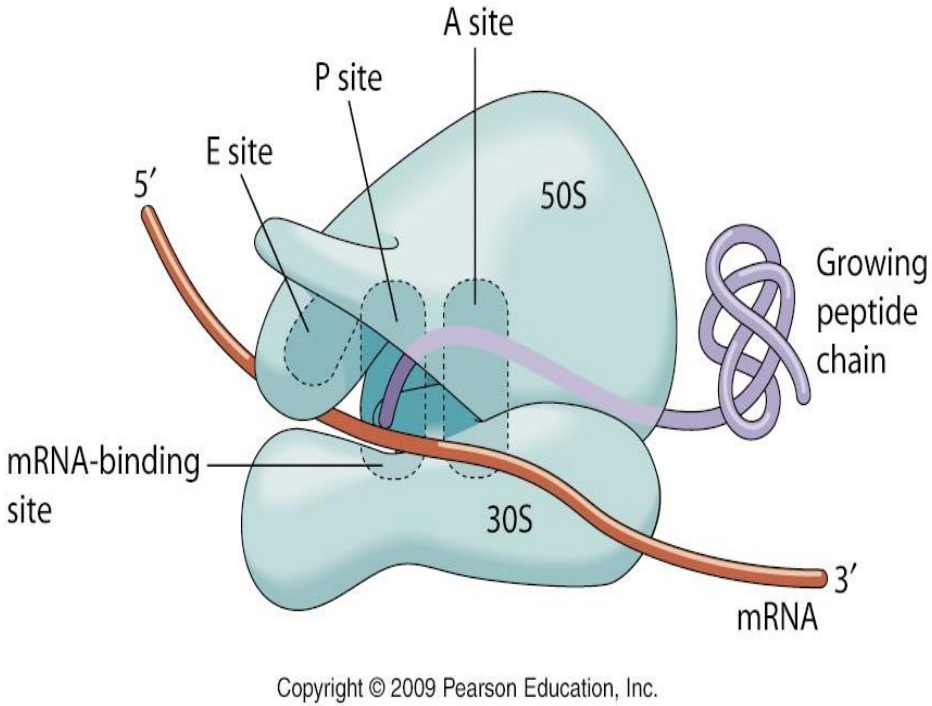
Eukaryotic Profiles

- Plasmids
  - –Not found in Animal cells
  - –May be found in Yeast and plant cells

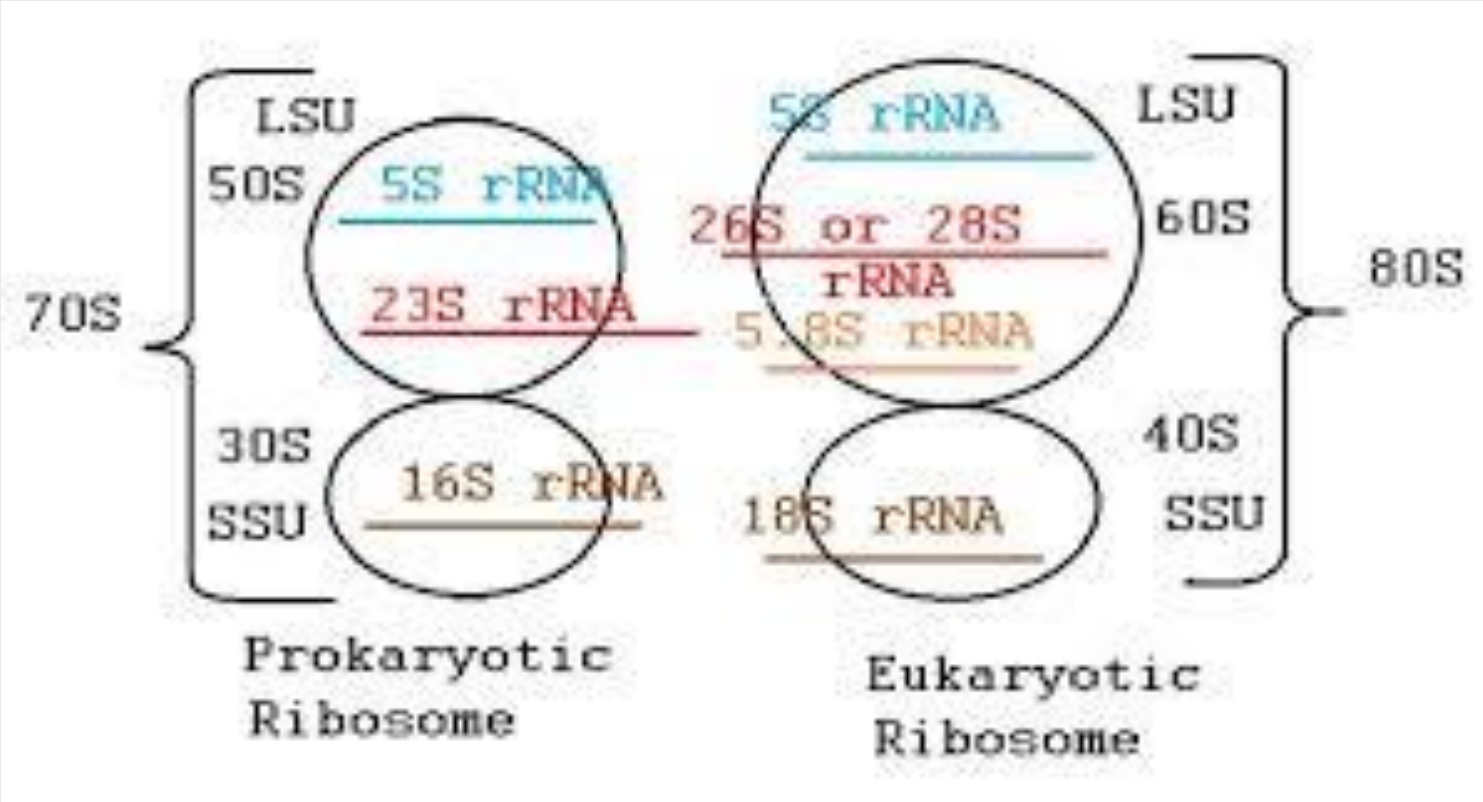
Prokaryotic Profiles

Eukaryotic Profiles

- Ribosomes
  - Made of 60% ribosomal RNA and 40% protein-Consist of two subunits: large and small- Prokaryotes have three ribosomal RNAs (23S, 16S, and 5S), which are equivalent to the 28S, 18S, and 5S rRNAs of eukaryotic cells .
  - eukaryotic ribosomes in size and number of proteins
  - Sites of protein synthesis



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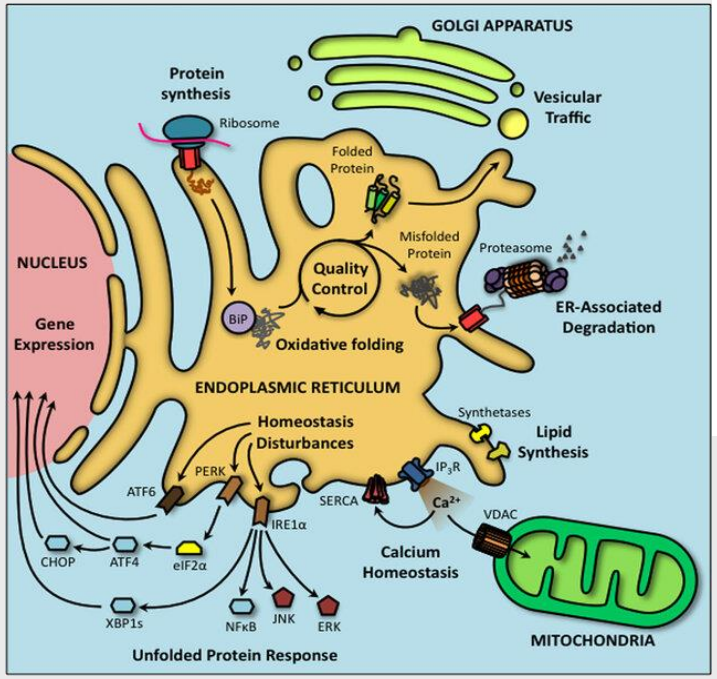
Prokaryotic Profiles

No Endoplasmic Reticulum

Eukaryotic Profiles

Endoplasmic Reticulum

It is a network of passages connected to the plasma membrane and the nucleus. Its function is to transport materials between the cytoplasm and the nucleus. It is the center of vital processes in the cell. It has two types: Rough endoplasmic reticulum: ribosomes spread on its surface. Smooth endoplasmic reticulum: ribosomes are not scattered on its surface. It is the center of vital processes in the cell, so it appears more widespread.

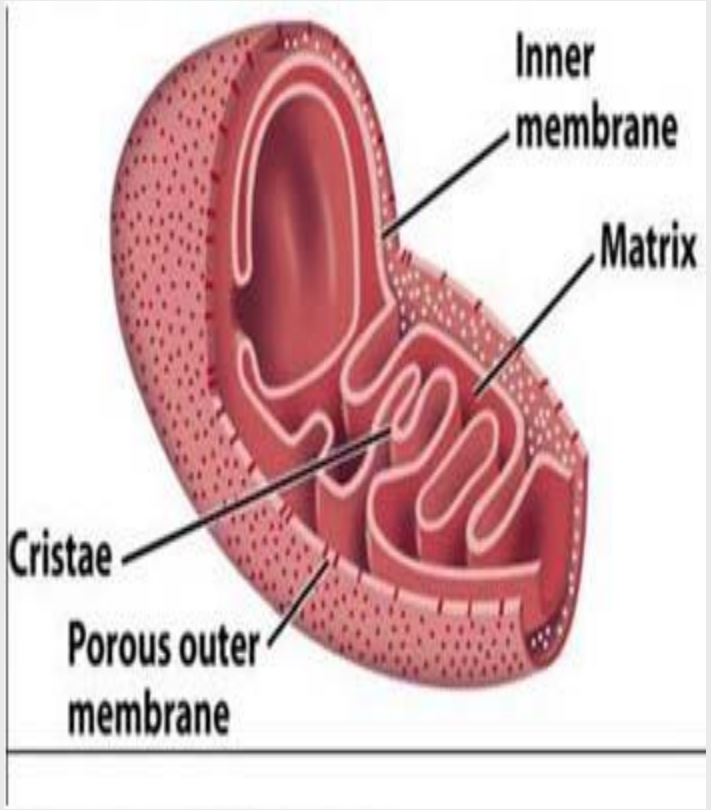


Prokaryotic Profiles

Eukaryotic Profiles

No Mitochondria

Mitochondria is called the energy house



A cylindrical organelle consisting of two membranes (smooth outer and inner (containing folds)

Mitochondria are surrounded by a double membrane Inner membrane: branches into folds called the cristae and the inner cavity contains an interstitial substance called the **matrix**.

It contains enzymes that play a significant role in the process of the vital activity of fungi related to demolition and construction, i.e. breaking and decomposing food and releasing energy from it, and.

It contains a DNA called Genomic mitochondrial DNA because it carries a genetic code for the production of proteins essential for the work of mitochondria

Prokaryotic Profiles

No Golgi bodies



Eukaryotic Profiles

Golgi bodies

- They are called dicteosomes, consisting of sacs arranged in a parallel arrangement and of spherical vesicles.
- consists of proteins and glycoprotein.
- It is responsible for the formation of the new cell walls
- also serve to store, prepare and export materials from the cell.

Prokaryotic Profiles

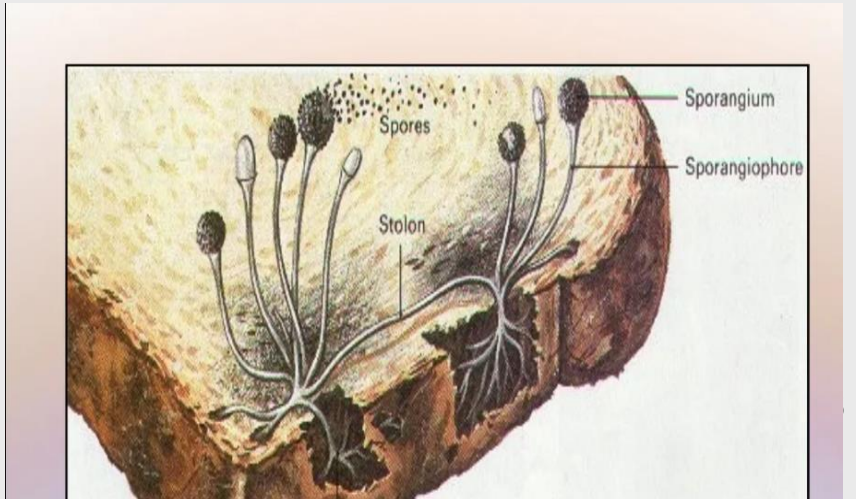
Motility By flagella

- 3 parts:
  - –**Filament** – long, thin, helical structure composed of protein **Flagellin**
  - –**Hook** – curved sheath
  - –**Basal body** – stack of rings firmly anchored in cell wall
- Rotates 360º
- Functions in motility of cell through environment

Eukaryotic Profiles

Motility

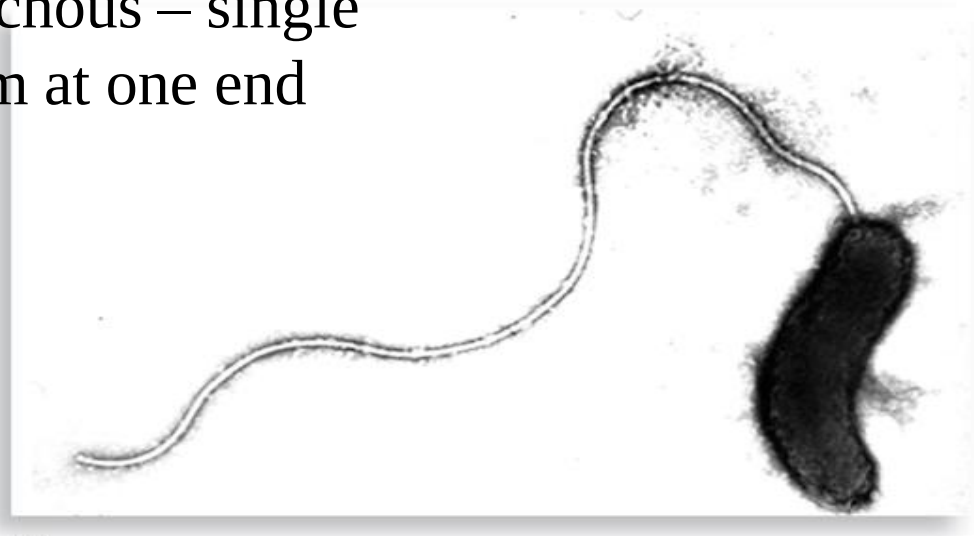
Fungi can't move around so they make spores that are like seeds. Spores fly away on the breeze or in water, on animals or clothing and find a new place to grow that has everything they need. If they can't find one, they just hibernate - they sleep until the right place comes along



Flagella Arrangements

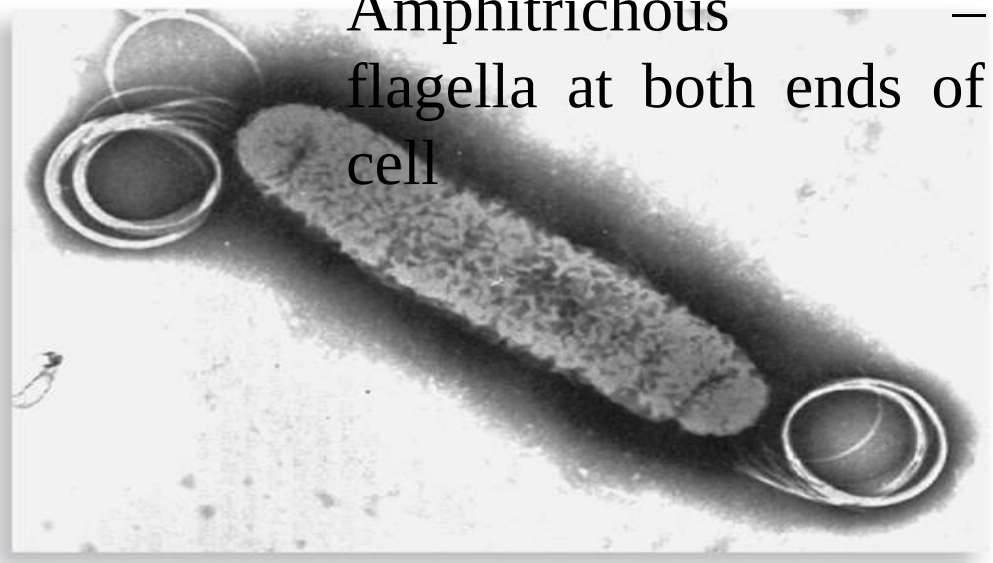
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Monotrichous – single flagellum at one end



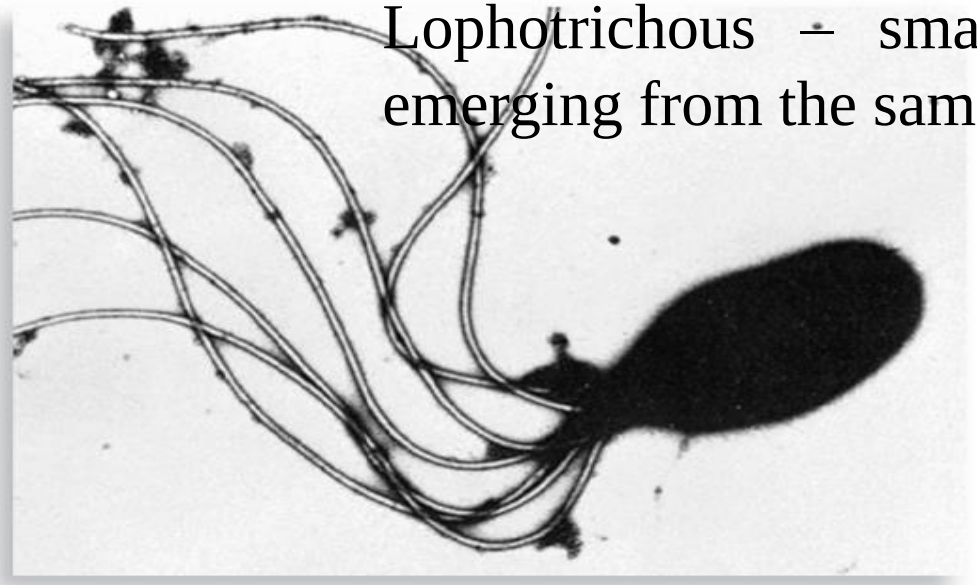
(a)

Amphitrichous – flagella at both ends of cell



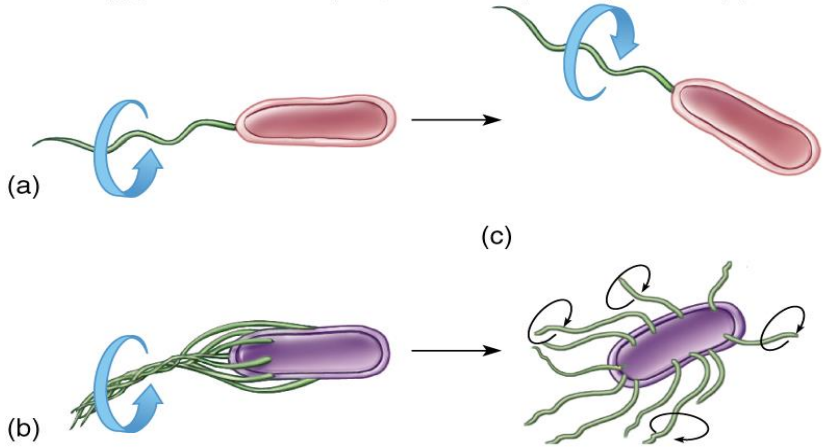
(c)

Lophotrichous – small bunches emerging from the same site



(b)

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(a)

(c)

(b)

Prokaryotic Profiles

Motility By flagella

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Eukaryotic Profiles

Motility

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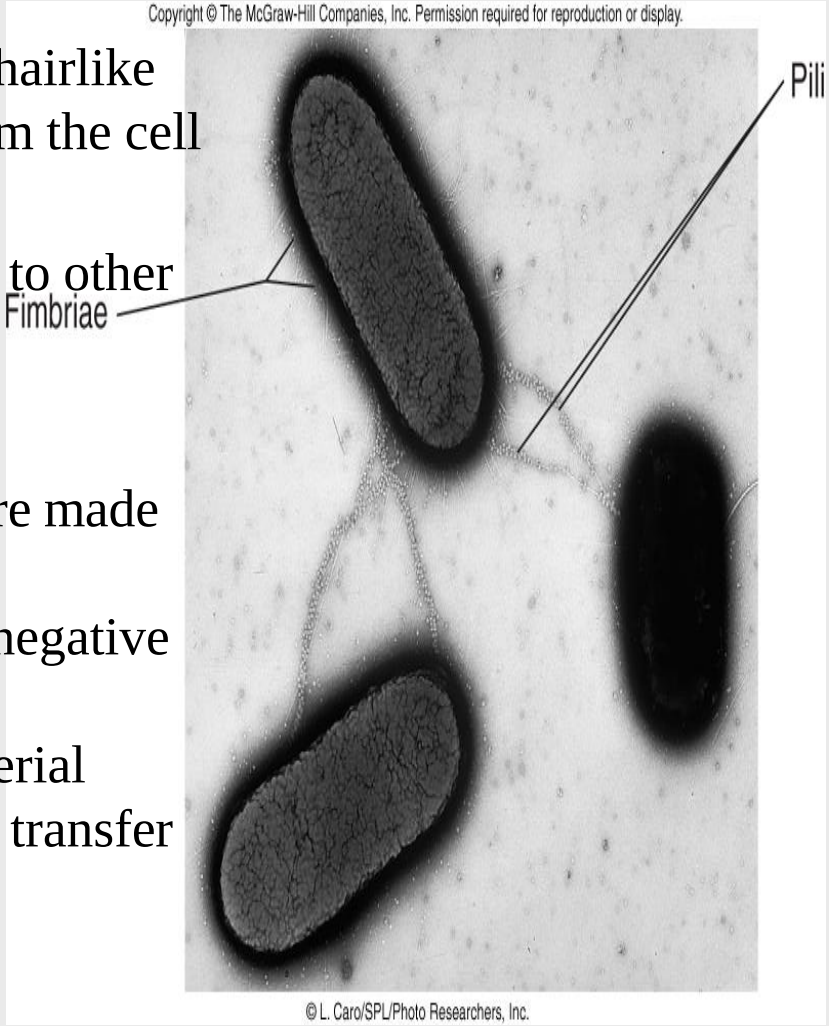


Prokaryotic Profiles

Attachment by

- **Fimbriae**
  Fine, proteinaceous, hairlike bristles emerging from the cell surface
- Function in adhesion to other cells and surfaces

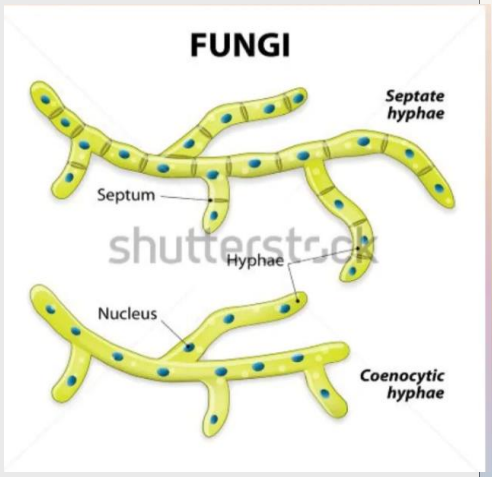
- **Pili**
- Rigid tubular structure made of **pilin** protein
- Found only in gram-negative cells
- Function to join bacterial cells for partial DNA transfer called **conjugation**

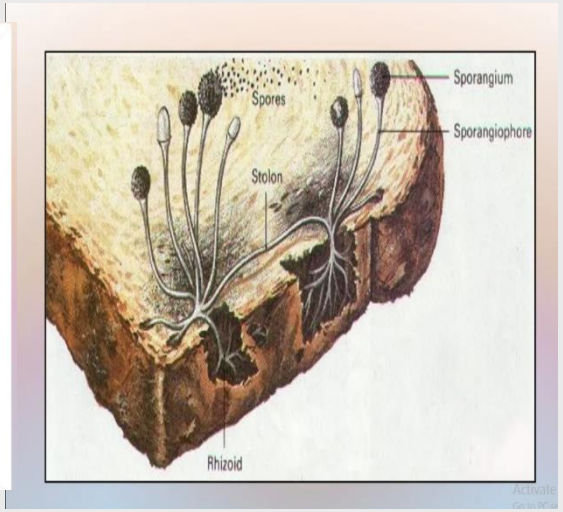


Eukaryotic Profiles

Attachment by

- **Growing mycelium**
- Mucilaginous substances





Mycelium arises from the germination as a any somatic cell Tube germination and elongates with the growing top, increases in thickness and branches by developing apex

Prokaryotic Profiles

Intracellular storage

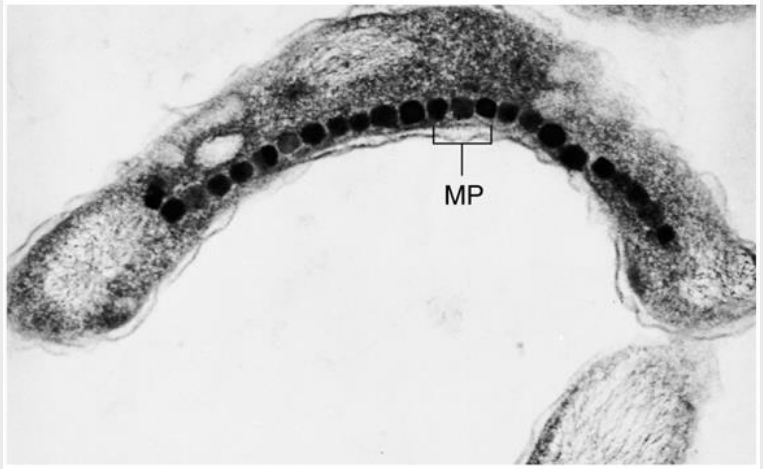
Examples: poly β-hydroxybutyrate, gas vesicles, sulfur and phosphate granules, particles of iron oxide

use them when environmental sources are depleted

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(a)



(b)

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Eukaryotic Profiles

Intracellular storage

Examples: glycogen,, gas vesicles, sulfur and phosphate granules, particles of iron oxide

use them when environmental sources are depleted